WEEK 4 - NEURAL NETWORKS FOR NLP

Revision Questions with Indicative Marking Scheme

Suggested total: 100 marks. The marking scheme is indicative and rewards technically correct explanations using the concepts covered in Week 4.

1. What is an artificial neural network? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: A computational model made of interconnected artificial neurons arranged in layers that learns patterns from data.

2. What is the role of an artificial neuron? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: It receives inputs, applies weights and bias, computes a weighted sum, then applies an activation function.

3. What are inputs, weights and bias in a neuron? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: Inputs are incoming feature values; weights control the importance of inputs; bias shifts the neuron's weighted sum.

4. Write the basic equation used to calculate the weighted sum of a neuron. [1 mark]

Marking: Award 1 mark for the essential idea. Expected: $z = \Sigma(w_i x_i) + b$.

5. What is an activation function, and why is it needed? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: A function applied to the weighted sum; it introduces non-linearity so the network can learn complex patterns.

6. Explain the purpose of ReLU. [1 mark]

Marking: Award 1 mark for the essential idea. Expected: ReLU outputs max(0,z): positive values pass through and negative values become zero.

7. What type of output does Sigmoid produce? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: A value between 0 and 1, commonly interpreted as a probability in binary classification.

8. Why is Softmax useful for multi-class text classification? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: It converts output scores into class probabilities that sum to 1.

9. Distinguish between input, hidden and output layers. [1 mark]

Marking: Award 1 mark for the essential idea. Expected: Input receives features; hidden layers learn intermediate representations/patterns; output produces the final prediction.

10. What happens inside a hidden layer? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: Weighted sums and activation functions transform inputs into learned intermediate features.

11. What is forward propagation? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: Information moves from input through the network to produce an output/prediction.

12. What is the difference between prediction and actual answer? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: Prediction is the model's output; actual answer/ground truth is the correct target label.

13. What is a loss function? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: A function measuring how far the model's prediction is from the correct answer.

14. Why minimise loss? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: Lower loss indicates predictions are closer to the correct targets.

15. What is backpropagation? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: The process of propagating error information backward to compute gradients for weights and biases.

16. What information does a gradient provide? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: It indicates how changing a parameter affects the loss and therefore the direction for adjustment.

17. What is gradient descent? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: An optimisation method that updates parameters in the direction that reduces loss.

18. What is the learning rate? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: The step size used when updating parameters during gradient descent.

19. What is an epoch? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: One complete pass through the entire training dataset. **20. What is a batch and batch size? [1 mark]**

Marking: Award 1 mark for the essential idea. Expected: A batch is a subset of training examples processed before an update; batch size is the number of examples in that subset.

21. Distinguish training, validation and test data. [1 mark]

Marking: Award 1 mark for the essential idea. Expected: Training learns parameters; validation supports tuning/model decisions; test evaluates final performance on unseen data.

22. What is underfitting? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: The model is too simple or insufficiently trained to capture the underlying patterns.

23. What is overfitting? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: The model learns the training data/noise too closely and performs poorly on unseen data.

24. What is dropout? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: A regularisation technique that randomly disables a fraction of neurons during training to reduce overfitting.

25. Why are word embeddings suitable neural-network inputs? [1 mark]

Marking: Award 1 mark for the essential idea. Expected: They represent words as dense numerical vectors that can encode useful semantic relationships.

Q1. Explain: Input ->

Weighted Sum -> Activation -> Hidden Layer(s) -> Output -> Prediction. [4 marks]

Marking scheme: 1 mark: inputs/features enter network; 1: weighted sum using weights and bias; 1: activation/hidden layers transform information; 1: output layer produces prediction/probabilities.

Q2. Explain how a neural network learns from an incorrect prediction: Prediction -> Loss -> Backpropagation -> Gradient Descent -> Weight Update. [4 marks]

Marking scheme: 1: prediction compared with true label/loss calculated; 1: backpropagation computes gradients; 1: gradient descent uses gradients and learning rate; 1: weights/biases updated to reduce future loss.

Q3. Explain forward propagation vs backpropagation. [3 marks]

Marking scheme: 1: forward pass moves input to output and produces prediction; 1: loss is calculated after prediction; 1: backpropagation moves learning/error information backward to compute gradients.

Q4. Probabilities: Positive 0.72, Neutral 0.18, Negative 0.10. What is predicted and why? [2 marks]

Marking scheme: 1: Positive; 1: because 0.72 is the highest output probability.

Q5. Why can't a neural network directly process 'This movie is excellent'? How can embeddings be used? [3 marks]

Marking scheme: 1: neural networks operate on numerical values; 1: tokenize/map words to embeddings; 1: dense vectors are supplied as numerical network inputs.

Q6. Excellent training performance but poor performance on new text: identify the problem and give three remedies. [4 marks]

Marking scheme: 1: overfitting; 3 marks: any three appropriate Week 4 strategies, e.g. dropout, more data, simpler model, regularisation, early stopping, data augmentation, reduced feature complexity.

Q7. Weighted Sum [4 marks]

Given x1=2, x2=3, w1=0.5, w2=0.2, b=0.1.
z=(0.5×2)+(0.2×3)+0.1 = **1.7**. ReLU(1.7)=**1.7**.
Marks: substitution 1; weighted products 1; z=1.7 1; ReLU output 1.

Q8. ReLU [3 marks]

ReLU(-3)=**0**; ReLU(0)=**0**; ReLU(2.5)=**2.5**. Negative values are set to zero.
Marks: 1 mark for each correct output; explanation may be used to support the first result.

Q9. Loss [4 marks]

L=-ln(0.8) ≈ **0.223**. If P(Positive) increases to 0.95, loss becomes **smaller** (about 0.051), because the model assigns greater probability to the correct class.
Marks: formula/substitution 1; 0.223 approx. 1; smaller 1; explanation 1.

Q10. Gradient Descent [4 marks]

w_new = 0.50 - (0.10×0.40) = **0.46**.
Marks: correct update rule 1; substitution 1; multiplication 0.04 1; final weight 0.46 1.

Q11. Epochs and Batches [5 marks]

1,000 / 100 = **10 batches per epoch**; therefore **10 updates per epoch**. Over 20 epochs: 10×20 = **200 updates**. **Marks:** correct batch calculation 2; updates per epoch 1; total updates 2.

Q12. Sentiment Classification [6 marks]

Marking scheme: Award 1 mark each for a coherent sequence covering: text/preprocessing or tokenisation; embedding representation; neural-network input; forward propagation through layers; Softmax/class

probabilities; class with highest probability selected as prediction.

Q13. Training the Model [7 marks]

Marking scheme: 1 mark each for: actual/true label; model prediction; loss compares prediction with target; backpropagation; gradients show parameter contribution/direction; learning rate controls update size; weights/biases updated to reduce loss.

Q14. Diagnose the Model [6 marks]

Marking scheme: 1: identifies overfitting; 2: explains very high training accuracy but substantially lower validation/test accuracy as poor generalisation; 3: one mark each for three suitable remedies such as dropout, early stopping, simpler model, more data, L1/L2 regularisation or data augmentation.

Q15. Training vs Validation vs Test [5 marks]

Marking scheme: 2: repeated test-set use influences model decisions and compromises its role as an unbiased final evaluation; 2: validation set should be used for tuning/early stopping; 1: test set should remain unseen until final evaluation.

Q16. Dropout [6 marks]

Marking scheme: 2: randomly disables/sets to zero a fraction of neurons during training; 1: different subsets can be active across training; 1: reduces reliance/co-adaptation; 1: acts as regularisation/reduces overfitting; 1: improves generalisation to unseen text.

Students should be able to draw and correctly label: (1) input-hidden-output layers; (2) forward propagation direction; (3) backpropagation direction; (4) the training cycle; and (5) underfitting/good fit/overfitting. **Award 1 mark for each recognisable and correctly labelled diagram/concept.**

A	Short questions	25
B	Understanding & explanation	20
C	Calculations	20
D	Application / exam-style	30
E	Diagram skills	5
	TOTAL	**100**

Exam preparation priority: Students should particularly practise forward propagation, loss, backpropagation, gradient descent, the complete text-classification pipeline, data splitting, and diagnosing overfitting.